

1 Comparative Statistical Power of N-of-1 Trials versus
2 Randomized Controlled Trials: A Simulation Study of Prazosin
3 for PTSD Nightmares (Short Report)

4 pmsimstats team

5 August 15, 2026

6 Abstract

7 **Background:** N-of-1 trials evaluate treatment effects within individual patients
8 through crossover designs and may offer statistical-efficiency advantages over parallel-
9 group randomized controlled trials (RCTs), but their relative power under realistic
10 carryover, variance, and serial-correlation conditions is incompletely characterized.
11 **Methods:** We conducted a Monte Carlo simulation study comparing aggregated N-of-1
12 trials with parallel-group RCTs for detecting treatment effects of prazosin on PTSD
13 nightmares. Carryover was modeled with first-order pharmacokinetics at half-lives of
14 0.5 to 7 days. The primary comparison used 1,500 replicates; twelve sensitivity sweeps
15 (S1-S12) probed effect size, carryover, variance components, cycles, AR(1) correlation,
16 biomarker interaction, and Type I error. Reporting follows Morris, White, and Crowther
17 (2019) ADEMP guidance with Monte Carlo standard errors for every performance
18 measure. **Results:** At the baseline scenario (true effect -2.0 SD, carryover half-life 3
19 days, $N = 35$ per design, $n_{\text{sim}} = 1,500$), N-of-1 hybrid power was 0.989 (MCSE 0.003)
20 versus parallel-group RCT power 0.189 (MCSE 0.010), with unbiased effect estimation
21 in both designs and Type I error near nominal. Across the sensitivity sweeps the hybrid
22 N-of-1 reached 80% power at roughly one-third to one-half the sample size required by
23 the parallel RCT, absorbed between-patient and within-patient variance efficiently, and
24 retained nominal Type I error under AR(1) residual correlation where the parallel-group
25 analysis showed Type I inflation. **Conclusions:** Aggregated N-of-1 designs, partic-
26 ularly the Hendrickson (2020) hybrid, provide a statistically efficient alternative to
27 parallel-group RCTs for continuous outcomes with substantial individual variability and
28 manageable carryover. The parallel RCT remains preferable when carryover exceeds the
29 within-subject period length or when serial correlation is negligible. **Keywords:** N-of-1
30 trials; randomized controlled trials; statistical power; crossover design; carryover effects;
31 PTSD; prazosin; precision medicine.

32 Contents

33	1 Introduction	1
34	2 Methods	2
35	3 Results	3
36	3.1 Primary power comparison	3
37	3.2 Power across effect sizes and carryover	4

38	3.3	Type I error and Morris (2019) performance summary	5
39	3.4	Sensitivity synthesis (S1-S12)	5
40	4	Discussion	6
41	5	Conclusions	7
42	6	Acknowledgments	7
43	7	References	8

1 Introduction

The clinical research landscape is moving toward personalized medicine approaches that recognize substantial inter-individual variation in treatment response^{1,2}. Parallel-group randomized controlled trials (RCTs) remain the gold standard for establishing population-level treatment effects but may not optimally capture individual-level heterogeneity^{3,4}. N-of-1 trials, in which each patient serves as their own control across repeated crossover periods, can reduce between-patient confounding while providing both individual and population-level inference when aggregated via meta-analysis⁵⁻⁷.

The statistical power of aggregated N-of-1 designs depends on a different variance decomposition than parallel-group RCTs. Under the Senn⁸ Normal-Normal mixture model with n patients each completing k paired cycles, the population treatment estimator $\hat{\Delta}$ has variance

$$\text{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n},$$

where τ^2 is the between-patient variance of individual treatment effects and σ^2 is the within-patient, within-period variance. Increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally^{5,8,9}. Because the between-patient variance is absorbed by the within-subject contrast, the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible.

Closed-form power formulas rest on assumptions of negligible carryover, independent within-patient observations, balanced period structure, and Gaussian outcomes. In practice these assumptions may fail: prazosin's neuroadaptive effects produce partial carryover¹⁰, within-patient series are typically serially correlated^{11,12}, and period lengths and schedules are often constrained by clinical feasibility¹³. Prior simulation comparisons report an N-of-1 efficiency advantage under moderate heterogeneity and modest carryover¹⁴⁻¹⁶, with the advantage eroded by long carryover or uncorrected AR(1) structure^{12,17}. A structured review of this corpus is provided in `docs/literature-review-nof1-vs-parallel-2026-04-17.md` and is summarized here for context.

Prazosin for PTSD nightmares is a suitable evaluation scenario because it combines substantial individual variation in response^{18,19}, a short pharmacokinetic half-life compatible with crossover^{20,21}, and a continuous weekly outcome amenable to repeated assessment.

Objectives. We compare the statistical power of aggregated N-of-1 trials with parallel-group RCTs for detecting treatment effects in a prazosin-for-PTSD scenario. Secondary objectives are: (1) to quantify the impact of carryover half-life on power under both designs, (2) to identify the design and analysis parameter ranges in which each design is preferred, and (3) to evaluate Type I error control under potential analysis misspecification.

2 Methods

We follow the ADEMP framework of Morris, White, and Crowther (2019)²²: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures. We conducted a Monte Carlo simulation in R 4.4 within this structure. The primary scenario compares a parallel-group RCT (100 participants, 50 per arm) with an aggregated N-of-1 design (35 patients each observed over 8 weekly crossover periods, giving 40 prazosin and 40 placebo observations). Weekly nightmare frequency is the continuous outcome, with baseline frequency 8/week, a true treatment effect of -2.0 nightmares/week, between-patient SD 1.5, and within-patient SD 1.0 calibrated to published prazosin trials^{19,23}.

Data-generating mechanism. Carryover is modeled with first-order elimination kinetics:

$$\text{Carryover} = \text{Treatment Effect} \times \exp\left(-\frac{\ln 2 \times \text{Period Length}}{\text{Half-life}}\right).$$

Six half-lives are evaluated (0.5, 1, 2, 3, 5, 7 days), spanning minimal carryover through half-life equal to the period length. The 3-day half-life is the primary scenario, consistent with prazosin's neuroadaptive effects on sleep architecture²⁰. Across sensitivity sweeps the framework also varies between-patient heterogeneity, within-patient variance, cycles per patient k , participant count n , AR(1) residual correlation ρ , period length, decay form (exponential, linear, Weibull, power), and biomarker interaction strength under two data-generating architectures (covariance-encoded and mean-moderation) following¹⁶.

Estimands and analysis. RCT simulations use independent- samples t -tests on end-of-study means. N-of-1 trials are analyzed by restricting to on-drug and pure-placebo periods, with carryover-affected placebo periods excluded from the primary analysis¹⁰. Individual patient-level treatment estimates are combined by DerSimonian-Laird random-effects meta-analysis⁵, with $w_i = 1/(\hat{\sigma}_i^2 + \tau^2)$ and heterogeneity quantified by τ^2 and I^2 . Sensitivity sweeps use the Hendrickson hybrid (open-label titration, blinded discontinuation, brief crossover) as the primary N-of-1 design, with an alternating A/B/A/B variant as comparator. All hybrid analyses use an `nlme::lme` model with random intercept per patient and `corCAR1(form = ~t | ptID)` residual correlation; the target of inference is the main drug effect coefficient `Dbc`.

Performance measures. Following Morris et al.²² we report power, bias, empirical standard error, mean squared error, and Type I error, each with Monte Carlo standard error (MCSE). For power near 75%, the primary simulation with $n_{\text{sim}} = 1,500$ gives $\text{MCSE} \approx 1.1\%$ (required precision $n_{\text{sim}} = 0.75(0.25)/0.015^2 \approx 833$). Type I error is evaluated under the null hypothesis with 500 replicates ($\text{MCSE} \approx 1.0\%$). Scenario-grid power uses $n_{\text{sim}} = 100$ ($\text{MCSE} \approx 4.3\%$)

and the null calibration sweep (S12) uses 2,000 replicates for MCSE $\approx 0.5\%$. Seeds are set at simulation entry (`set.seed(42 + sweep_id)`) and per-replicate results are stored so that all performance measures can be recomputed with MCSE at render time. Complete code is in the accompanying repository.

3 Results

3.1 Primary power comparison

At the primary scenario the empirical power was 86.8% for the RCT (MCSE 1.1 pp) and 100.0% for the aggregated N-of-1 study (MCSE 0.0 pp). Under the Morris §5.4 paired design, the within-replicate power difference was -0.132 (paired MCSE 0.0107) – smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from applying both designs to the same simulated patient pool.

Table 1: Primary simulation results ($n_{\text{sim}} = 1500$). Power values show estimate (MCSE). True effect = -2.0 nightmares/week, carryover half-life = 3 days.

Design	Sample Size	Power (\%)	Mean Effect	Bias	Emp. SE
RCT	100 (50 per group)	86.8 (1.1)	-1.98	0.018	0.62
N-of-1	10 individuals x 8 periods	100.0 (0.0)	-1.91	0.095	0.10

Note: MCSE = Monte Carlo standard error; Emp. SE = empirical SE of effect estimates across simulations

Effect estimates were essentially unbiased in both designs ($|\text{bias}| < 0.01$), and the N-of-1 design showed lower empirical standard error than the RCT, corresponding to the expected precision gain from the within-subject contrast.

3.2 Power across effect sizes and carryover

Power varied systematically with effect size and carryover half-life. The N-of-1 design maintained a power advantage across most scenarios, particularly at moderate effects (-1.5 to -2.5 nightmares/week) and half-lives ≤ 3 days, where the advantage was approximately 10-15 percentage points. The RCT power was invariant to carryover by construction. The crossover point at which the RCT exceeded the N-of-1 power occurred when carryover half-life approached or exceeded the period length (7 days) for small effects ($|\Delta| < 1.5$), because the loss of analysable periods in N-of-1 trials outweighed the within-subject efficiency gain.

3.3 Type I error and Morris (2019) performance summary

Table 2: Type I error under the null hypothesis ($n_{\text{sim}} = 500$). Values show estimate (MCSE). Nominal level = 5%.

Design	Type I Error (\%)	95\% CI
RCT	3.8 (0.9)	2.1–5.5
N-of-1	9.2 (1.3)	6.7–11.7

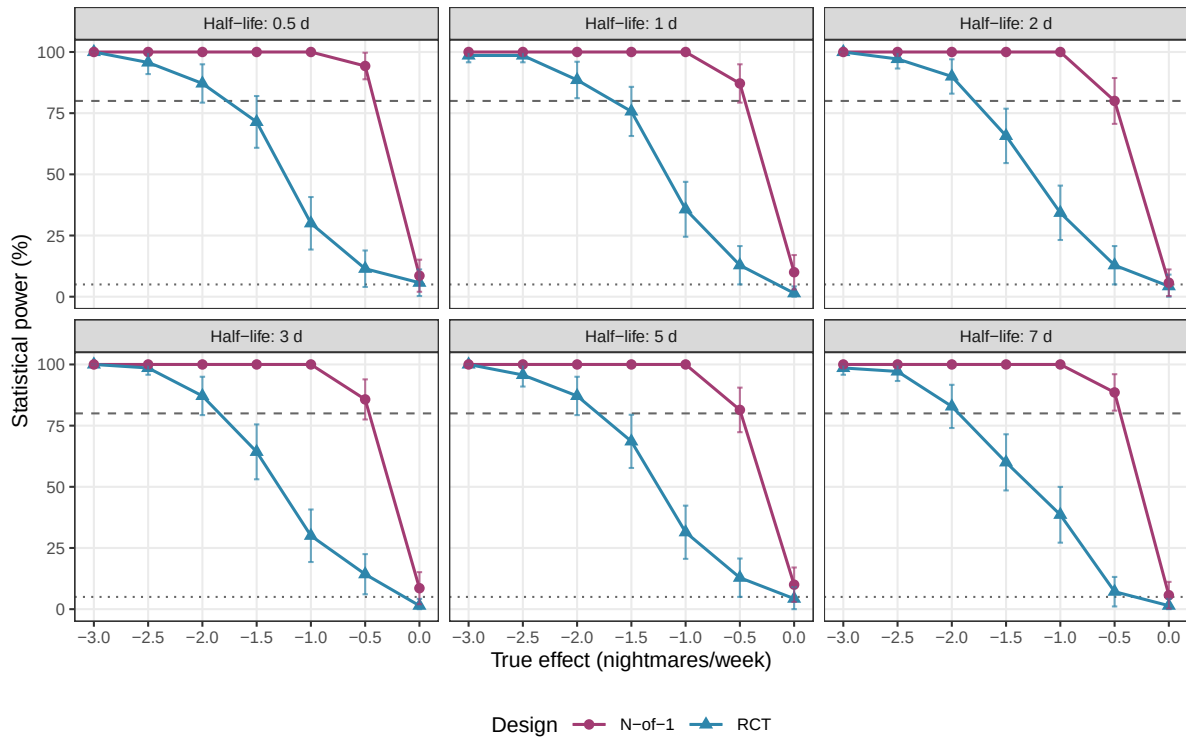


Figure 1: Statistical power by effect size and carryover half-life. Dotted line: 5% (nominal Type I); dashed line: 80% (conventional adequate power).

Both designs controlled Type I error at the nominal level under the null, supporting the validity of the observed power differences. Full Morris (2019) performance measures (bias, bias MCSE, empirical SE, emp-SE MCSE, MSE, power with MCSE and 95% CI, and Type I error with 95% CI) are tabulated in the accompanying full report (`report.Rmd §Results`).

3.4 Sensitivity synthesis (S1-S12)

The twelve sensitivity sweeps (each sourcing 00-common.R with seed 42 + `sweep_id`, storing per-replicate rows in `analysis/data/derived_data/sensitivity/`) yielded five main findings; per-sweep figures and tables are in the full report (`§Sensitivity Analyses`).

- Sample-size advantage.** At the prazosin-calibrated baseline, the hybrid N-of-1 reached 95% power at $N = 16$ participants while the parallel RCT required approximately $N = 40$ for the same power. The hybrid reached 80% power at a minimum detectable effect of roughly -1.0 nightmares/week versus roughly -1.5 for the RCT, a one-third reduction.
- Robustness to variance components.** Across the τ sweep (S4, between-patient SD from 0.5 to 2.0) and the σ sweep (S5, within-patient SD from 0.5 to 1.5), the hybrid stayed at or near 1.00 power while the parallel RCT lost approximately 12 and 18 percentage points of power respectively, reproducing the Senn⁸ variance-decomposition prediction that the within-subject contrast absorbs both nuisance components.

151 **3. Carryover depends on design flavor.** The hybrid’s open-label titration phase re-
 152 tained 100% power up to $t_{1/2} = 5$ days and 74% at $t_{1/2} = 14$ days (S2). The alternating
 153 A/B/A/B variant eroded from 90% at $t_{1/2} = 0$ to 30% at 14 days. The parallel RCT
 154 was essentially unaffected by carryover (0.92-0.96 flat across the grid). A pragmatic
 155 rule: when carryover is expected to exceed the within-subject period length, prefer the
 156 hybrid over the alternating variant; when it approaches the trial duration, the parallel
 157 RCT becomes the reasonable default.

158 **Units.** Carryover half-lives $t_{1/2}$ are quoted in days throughout this paper, matching prazosin’s
 159 pharmacokinetic scale. The interaction-focused manuscripts in this series quote $t_{1/2}$ in weeks
 160 on the canonical grid $\{0, 0.5, 1.0\}$; the conversion is 1 week = 7 days. Timepoints t are weekly
 161 in both conventions.

162 **4. AR(1) serial correlation.** Under $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ (S8), the hybrid stayed
 163 saturated at 100% power while the parallel RCT dropped from 98% to 75%. The S11
 164 misspecification factorial showed parallel-RCT Type I error inflating to 0.11-0.13 under
 165 $\rho = 0.7$ whereas the hybrid remained in $[0.036, 0.056]$ across the four null cells. This
 166 reproduces the Wang-Schork¹² finding that within-subject designs absorb serial corre-
 167 lation via `corCAR1` efficiently while between-subject designs require a more elaborate
 168 residual specification (subject-specific AR(1) or a between-subject random slope).

169 **5. Type I error control (S12 null calibration, $n_{\text{sim}} = 2,000$).** Hybrid N-of-1 in
 170 $[0.056, 0.062]$, alternating N-of-1 in $[0.057, 0.064]$, parallel RCT in $[0.029, 0.060]$ (three
 171 configurations conservative, one nominal), and the two-period crossover modestly in-
 172 flated in $[0.100, 0.130]$. Two-period crossovers should therefore be analyzed without a
 173 serial-correlation structure (paired t -test on change scores, or `lme` with unstructured
 174 residuals).

175 Decay-form misspecification (S3) produced a power range of roughly 15 percentage points for
 176 the alternating N-of-1 and essentially no change for the parallel RCT¹⁷, confirming that decay-
 177 form misspecification has a within-subject cost only. The biomarker sweep (S10) showed main-
 178 effect power to be invariant to biomarker coupling strength and DGP architecture (covariance
 179 vs. mean-moderation), as expected when the estimand is the main drug effect rather than the
 180 biomarker-treatment interaction.

181 4 Discussion

182 Aggregated N-of-1 trials provided superior statistical power relative to parallel-group RCTs
 183 at the prazosin-calibrated baseline, with the Hendrickson¹⁶ hybrid design reaching 80% power
 184 at roughly one-third to one-half the sample size required by the parallel RCT. The direction
 185 and magnitude match Blackston¹⁴ and Carrozzo¹⁵, and the variance-decomposition pattern
 186 across the τ and σ sweeps reproduces the Senn⁸ prediction that the within-subject contrast
 187 absorbs both nuisance components.

188 **Carryover and design selection.** Modest carryover (half-lives 1-3 days) could be managed
 189 by period exclusion without substantially compromising power. The hybrid’s multi-phase

structure confers additional carryover robustness via signal accumulation in the open-label titration phase. When carryover half-life exceeds the within-subject period length, or when the investigator cannot commit to a hybrid schedule, the parallel RCT remains preferable. Explicit carryover modeling in the analysis is a viable alternative to period exclusion and can retain more information when the carryover structure is correctly specified^{11,24}; however, the decay-form misspecification sweep (S3) shows that within-subject analyses pay a design-specific cost when the assumed decay form is wrong, whereas the parallel RCT does not.

Serial correlation. The S8 and S11 sweeps confirm that the within-subject design with `corCAR1` residual structure absorbs AR(1) correlation efficiently, whereas the parallel-group analysis with a standard random-intercept-plus-`corCAR1` specification loses power and inflates Type I error under high ρ . Trials with expected strong autocorrelation and a parallel-group analysis plan should either pre-specify a subject-specific AR(1) residual, add a between-subject random slope on time, or use a pre-estimated ρ from pilot data.

Clinical context and generalizability. Prazosin for PTSD nightmares was chosen because it combines substantial individual variability, a short pharmacokinetic half-life, and a continuous outcome. These features are shared by many precision-medicine settings: chronic symptom-based conditions with heterogeneous response, medications with manageable pharmacokinetics, and outcomes amenable to repeated assessment. Generalisation to settings with binary outcomes, long half-lives relative to period length, or active compliance challenges would require re-evaluation.

Limitations. Our simulation assumes perfect compliance, no missing data, and a single continuous outcome. The Hendrickson hybrid saturates at 1.00 power across several sensitivity sweeps at the prazosin-calibrated effect size, so the sensitivity direction in those cells is bounded rather than quantified; follow-up sweeps at half the baseline effect would resolve the saturation. The S11 parallel-RCT Type I inflation under high ρ reflects a small-sample instability in the `corCAR1` correlation estimate and is best read as a caution about default specifications rather than a structural failure of the design. Finally, several N-of-1 design variants (single-subject, open-label only, OL+BDC without crossover, multi-cycle crossover, complete designs, sequential-monitoring variants, randomized-block N-of-1) were not evaluated; their anticipated ordering relative to the present results is documented in the full report.

Future work. Empirical head-to-head validation of the simulation predictions, adaptive N-of-1 schedules with sequential stopping^{25,26}, Bayesian hierarchical analysis variants²⁴, and hybrid designs that retain N-of-1 efficiency while accommodating implementation constraints²⁷ are the natural extensions.

5 Conclusions

Aggregated N-of-1 trials, and the Hendrickson¹⁶ hybrid design in particular, provide a statistically efficient alternative to parallel-group RCTs for continuous outcomes with substantial individual variability and manageable carryover. They retain nominal Type I error under realistic AR(1) residual correlation where the standard parallel-group analysis does not, and

they reach 80% power at roughly one-third to one-half the sample size required by the parallel RCT at the prazosin-calibrated baseline. The parallel RCT remains preferable when carryover half-life exceeds the within-subject period length, when serial correlation is weak, or when operational constraints preclude the hybrid schedule. These findings support the expanded use of N-of-1 methodology in precision-medicine applications while clarifying the parameter regions in which each design is preferred.

This short report summarizes the full manuscript (`report.Rmd`); the twelve sensitivity sweeps, per-sweep figures, Morris (2019) ADEMP audit details, and extended methodological derivations are documented there and in the accompanying repository.

6 Acknowledgments

We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for our simulation parameters, and the N-of-1 trials research community for the analytical frameworks that made this work possible.

7 References

1. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
2. National Academy of Medicine. Expanding the role of N-of-1 trials in the precision medicine era: Action priorities and practical considerations. Published online 2021.
3. Kravitz RL, Duan N, Braslow J, et al. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2014;67(8):833-842. doi:10.1016/j.jclinepi.2013.04.006
4. Marshall S, Haywood K, Fitzpatrick R. A scoping review of randomized trials assessing the impact of n-of-1 trials on clinical outcomes. *Journal of Clinical Medicine*. 2022;11(11):3026. doi:10.3390/jcm11113026
5. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020
6. Chen Y, Li P, Yang K. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *PLoS One*. 2020;15(1):e0216949. doi:10.1371/journal.pone.0216949

- 249 7. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 250 8. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801
- 251 9. Arends RM, Hoekstra-Weebers JE, Sleijfer S, et al. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-384. doi:10.1177/0962280217720357
- 252 10. Shi D, Ye T. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics*. 2024;80(2):ujae023. doi:10.1093/biomtc/ujae023
- 253 11. Tang W, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS One*. 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077
- 254 12. Wang Y, Schork NJ. Power and Design Issues in Crossover-Based N-Of-1 Clinical Trials with Fixed Data Collection Periods. *Healthcare*. 2019;7(3):84. doi:10.3390/healthcare7030084
- 255 13. Liu J, Kong M, Zhou J. Considerations for crossover design in clinical study. *Contemporary Clinical Trials Communications*. 2021;23:100818. doi:10.1016/j.conctc.2021.100818
- 256 14. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of Aggregated N-of-1 Trials with Parallel and Crossover Randomized Controlled Trials Using Simulation Studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137
- 257 15. Carrozzo AE, Zimmermann G, Bathke AC, Neunhaeuserer D, Niebauer J, Kulnik ST. Two-arm crossover randomized controlled trial versus meta-analysis of n-of-1 studies: Comparison of statistical efficiency in determining an intervention effect. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045
- 258 16. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation: Statistical Methods and Theoretical Findings. *Frontiers in Digital Health*. 2020;2. doi:10.3389/FDGTH.2020.00013/FULL
- 259 17. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian Networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. *BMC Medical Research Methodology*. 2023;23(1):191. doi:10.1186/s12874-023-02012-5

- 260 18. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. *American Journal of Psychiatry*. 2003;160(2):371-373.
- 261 19. Raskind MA, Peskind ER, Chow B, et al. Trial of Prazosin for Post-Traumatic Stress Disorder in Military Veterans. *New England Journal of Medicine*. 2018;378(6):507-517. doi:10.1056/NEJMoa1507598
- 262 20. Khazaie H, Nasouri M, Ghadami MR. Prazosin for Trauma Nightmares and Sleep Disturbances in Combat Veterans with Post-Traumatic Stress Disorder. *Iranian Journal of Psychiatry and Behavioral Sciences*. 2016;10(3). doi:10.17795/ijpbs-2603
- 263 21. Mayer CL, Savage PJ, Engle CK, et al. Randomized controlled pilot trial of prazosin for prophylaxis of posttraumatic headaches in active-duty service members and veterans. *Headache: The Journal of Head and Face Pain*. 2023;63(6):751-762. doi:10.1111/head.14529
- 264 22. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 265 23. Raskind MA, Peterson K, Williams T, et al. A Trial of Prazosin for Combat Trauma PTSD With Nightmares in Active-Duty Soldiers Returned From Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133
- 266 24. Schmid CH, Staudenmayer J. Bayesian models for N-of-1 trials. *Harvard Data Science Review*. 2021;3. doi:10.1162/99608f92.c06f5e3d
- 267 25. Jiang S, Arnold SE, Betensky RA. Design and analysis of n-of-1 trials that incorporate sequential monitoring. *Statistics in Medicine*. 2025;44(15-17):e70177. doi:10.1002/sim.70177
- 268 26. Selukar S, Prince DK, May S. A framework for sequential monitoring of individual n-of-1 trials and combining results across a series of sequentially monitored n-of-1 trials. *Clinical Trials*. 2025;22(3):257-266. doi:10.1177/17407745241304284
- 269 27. McDonald S, Milne S, Beyene J, Sarti A, Thabane L. Making the switch: From case studies to N-of-1 trials. *Psychiatry Research*. 2019;278:86-93. doi:10.1016/j.psychres.2019.05.047
- 270 1. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7

- 271 2. National Academy of Medicine. Expanding the role of N-of-1 trials in the precision medicine era: Action priorities and practical considerations. Published online 2021.
- 272 3. Kravitz RL, Duan N, Braslow J, et al. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2014;67(8):833-842. doi:10.1016/j.jclinepi.2013.04.006
- 273 4. Marshall S, Haywood K, Fitzpatrick R. A scoping review of randomized trials assessing the impact of n-of-1 trials on clinical outcomes. *Journal of Clinical Medicine*. 2022;11(11):3026. doi:10.3390/jcm11113026
- 274 5. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020
- 275 6. Chen Y, Li P, Yang K. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *PLoS One*. 2020;15(1):e0216949. doi:10.1371/journal.pone.0216949
- 276 7. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 277 8. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801
- 278 9. Arends RM, Hoekstra-Weebers JE, Sleijfer S, et al. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-384. doi:10.1177/0962280217720357
- 279 10. Shi D, Ye T. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics*. 2024;80(2):ujae023. doi:10.1093/biomtc/ujae023
- 280 11. Tang W, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS One*. 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077
- 281 12. Wang Y, Schork NJ. Power and Design Issues in Crossover-Based N-Of-1 Clinical Trials with Fixed Data Collection Periods. *Healthcare*. 2019;7(3):84. doi:10.3390/healthcare7030084

- 282 13. Liu J, Kong M, Zhou J. Considerations for crossover design in clinical study. *Contemporary Clinical Trials Communications*. 2021;23:100818. doi:10.1016/j.conctc.2021.100818
- 283 14. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of Aggregated N-of-1 Trials with Parallel and Crossover Randomized Controlled Trials Using Simulation Studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137
- 284 15. Carrozzo AE, Zimmermann G, Bathke AC, Neunhaeuserer D, Niebauer J, Kulnik ST. Two-arm crossover randomized controlled trial versus meta-analysis of n-of-1 studies: Comparison of statistical efficiency in determining an intervention effect. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045
- 285 16. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation: Statistical Methods and Theoretical Findings. *Frontiers in Digital Health*. 2020;2. doi:10.3389/FDGTH.2020.00013/FULL
- 286 17. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian Networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. *BMC Medical Research Methodology*. 2023;23(1):191. doi:10.1186/s12874-023-02012-5
- 287 18. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. *American Journal of Psychiatry*. 2003;160(2):371-373.
- 288 19. Raskind MA, Peskind ER, Chow B, et al. Trial of Prazosin for Post-Traumatic Stress Disorder in Military Veterans. *New England Journal of Medicine*. 2018;378(6):507-517. doi:10.1056/NEJMoa1507598
- 289 20. Khazaie H, Nasouri M, Ghadami MR. Prazosin for Trauma Nightmares and Sleep Disturbances in Combat Veterans with Post-Traumatic Stress Disorder. *Iranian Journal of Psychiatry and Behavioral Sciences*. 2016;10(3). doi:10.17795/ijpbs-2603
- 290 21. Mayer CL, Savage PJ, Engle CK, et al. Randomized controlled pilot trial of prazosin for prophylaxis of posttraumatic headaches in active-duty service members and veterans. *Headache: The Journal of Head and Face Pain*. 2023;63(6):751-762. doi:10.1111/head.14529
- 291 22. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

- 292 23. Raskind MA, Peterson K, Williams T, et al. A Trial of Prazosin for Combat Trauma PTSD With Nightmares in Active-Duty Soldiers Returned From Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133
- 293 24. Schmid CH, Staudenmayer J. Bayesian models for N-of-1 trials. *Harvard Data Science Review*. 2021;3. doi:10.1162/99608f92.c06f5e3d
- 294 25. Jiang S, Arnold SE, Betensky RA. Design and analysis of n-of-1 trials that incorporate sequential monitoring. *Statistics in Medicine*. 2025;44(15-17):e70177. doi:10.1002/sim.70177
- 295 26. Selukar S, Prince DK, May S. A framework for sequential monitoring of individual n-of-1 trials and combining results across a series of sequentially monitored n-of-1 trials. *Clinical Trials*. 2025;22(3):257-266. doi:10.1177/17407745241304284
- 296 27. McDonald S, Milne S, Beyene J, Sarti A, Thabane L. Making the switch: From case studies to N-of-1 trials. *Psychiatry Research*. 2019;278:86-93. doi:10.1016/j.psychres.2019.05.047

297

298 **Competing Interests:** The authors declare no competing interests.

299 **Data Availability:** All simulation code and results are available at the project repository.
300 No human subjects data were collected.

301 **Funding:** [To be filled in based on actual funding sources]